higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

PROCESS CHEMISTRY
NQF LEVEL 4

(6050444)

1 March 2019 (Y-Paper)
13:00–16:00

This question paper consists of 7 pages.

NCV

TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

NC(V)

QUESTION 1

Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (1.1–1.5) in the ANSWER BOOK.

1.1 Linear Low-Density Polyethylene (LLDPE) is produced in gas phase reactors at low pressure and temperature.

1.2 Amplats situated in Machadodorp area of Mpumalanga is the world leader in the production of platinum.

1.3 Ammonia is a constituent in household cleaning products.

1.4 Asphalt is a sticky, black, and highly viscous liquid that is used for waterproofing roofs.

1.5 Fermentation temperature during the production of penicillin is 25-30 °C.

(5 × 1) **[5]**

QUESTION 2

Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–H) next to the question number (2.1–2.5) in the ANSWER BOOK.

COLUMN A		COLUMN B	
2.1	Materials that can be decomposed by living organisms	A	propane
2.2	A technique which converts metals into soluble salts in aqueous solution	B	hops
2.3	9-Azafluorene	C	biodegradable
2.4	Hydrocarbon with a chemical formula of C_3H_6	D	eluviation
2.5	A dried flower that is used to add bitterness to beer	E	leaching
		F	carbazole
		G	barley
		H	propylene
		I	carbazole alkaloids
		J	imperishable

(5 × 1) **[5]**

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QUESTION 3

3.1 List FOUR environmental health effects associated with discarded polymers. (4)

3.2 Polyethylene is a versatile synthetic resin used to manufacture many ordinary household items.

3.2.1 Name TWO properties of polyethylene. (2)

3.2.2 Differentiate between low-density polyethylene (LDPE) and high-density polyethylene (HDPE). (4)

3.3 Mixtures of some reagents are reacted to form PVC.

3.3.1 List THREE ingredients used in the production of PVC using a suspension polymerisation method. (3)

3.3.2 What are the TWO steps used to produce PVC? (2)

3.4 Explain the processing steps of solution polymerisation for SBR production. (6)

3.5 State THREE applications of SBR. (3)

3.6 Give FOUR examples of additives used in plastic production. (4)
[28]

QUESTION 4

4.1 Name FOUR producers (manufacturers) of base metals in South Africa. (4)

4.2 Which method (process) will you use to separate nickel, copper and cobalt from PGMs? (1)

4.3 Numerous ore processing methods are used in the production of base metals.

Sketch a labelled process flow diagram for the production of a mineral concentrate. (8)

4.4 South Africa together with Canada and Russia are the main producers of PGMs.

Name a precious metal that is used for each of the following:

4.4.1 Making of jewellery

4.4.2 Production of electronics, including computers, mobile phones and televisions

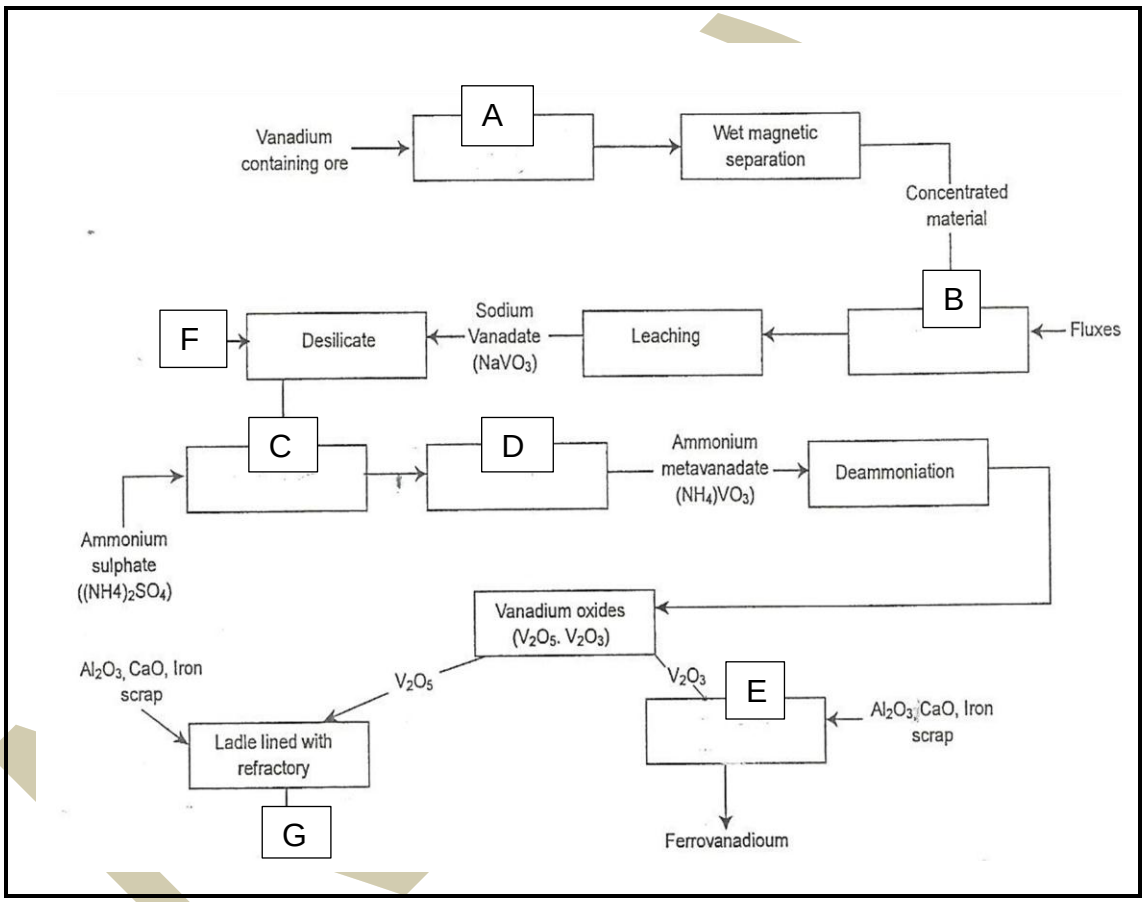
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4.4.3 A catalyst used for manufacturing processes and in catalytic converters

(3 × 1) (3)

4.5 Ferrovanadium is produced by aluminothermic or silicothermic vanadium containing different raw materials such as slag, spent catalysts and oil residues.

Complete the diagram below by naming the processes and products labelled (A–G) in the ANSWER BOOK.



(7)
[23]

QUESTION 5

5.1 Name the largest coal based chemicals company which was started in 1952 by the South African government. (1)

5.2 What are the THREE main uses of the following coal based products?

5.2.1 Ammonia

5.2.2 Hydrogen sulphide

(2 × 3) (6)

5.3 Chevron South Africa markets its own product under the brand name Caltex.

List FOUR products produced by Chevron. (4)

5.4 Name TWO companies that started Sapref in the 1960s. (2)

5.5 Crude oils are being classified as sweet or sour.

What is the difference between sweet and sour oil? (2)

5.6 Define the following cracking processes:

5.6.1 Hydrocracking

5.6.2 Thermal cracking

5.6.3 Catalytic cracking

(3 × 2) (6)

5.7 What was the primary disadvantage of the first thermal cracking method that was developed in 1913? (1)

5.8 State FOUR advantages of catalytic cracking over thermal cracking. (4)

[26]

QUESTION 6

6.1 Describe the functionality of the following biochemicals:

6.1.1 Enzymes

6.1.2 Hormones

(2 × 2) (4)

6.2 Name TWO manufacturers of pharmaceutical products in South Africa. (2)

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6.3 Discuss lautering in beer brewing. (4)

6.4 Explain the necessity of a cooling step in the downstream processing and purification of penicillin. (2)

6.5 What is a *nutrient broth*? (1)
[13]

TOTAL: 100

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